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Review Article

Role of Probiotic *Lactobacillus reuteri* in Improving Gut Health and Immunity in Infants and Toddlers: A Review

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ABSTRACT

The infant gut undergoes significant developmental phases that are fully dependent upon the colonization with microorganisms, beginning at birth. This colonization of gut microbiota has a powerful impact on host metabolic and immune homoeostasis. Probiotics strains exhibit a wide range of health benefits by modifying the intestinal microbiota and immunity. *Lactobacillus reuteri* is one of the most extensively studied probiotic strains. It promotes gut health by stimulation of mucosal gut barrier functions, production of antimicrobial substances (such as reuterin and lactic acid) and influencing acquired and innate immune responses. Reuterin produced by *L. reuteri* is a potent anti-microbial compound capable of inhibiting a wide spectrum of pathogenic microorganisms. Apart from antimicrobial metabolite production, *L. reuteri* creates biofilms that stimulate tumor necrosis factor production by lipopolysaccharide (LPS)- activated monocytoid cells. Interestingly, *L. reuteri* administration has emerged as a potential therapy for childhood functional gastrointestinal disturbances as these disturbances are associated with gut microbiota perturbations in early life. The current review summarizes the beneficial aspects of the probiotic *L. reuteri* strain in clinical practice with a special focus on its role in improving gut health and immunity in infants and toddlers.

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